
Responsiveness

Responsive Web Design Topics

What RWD is

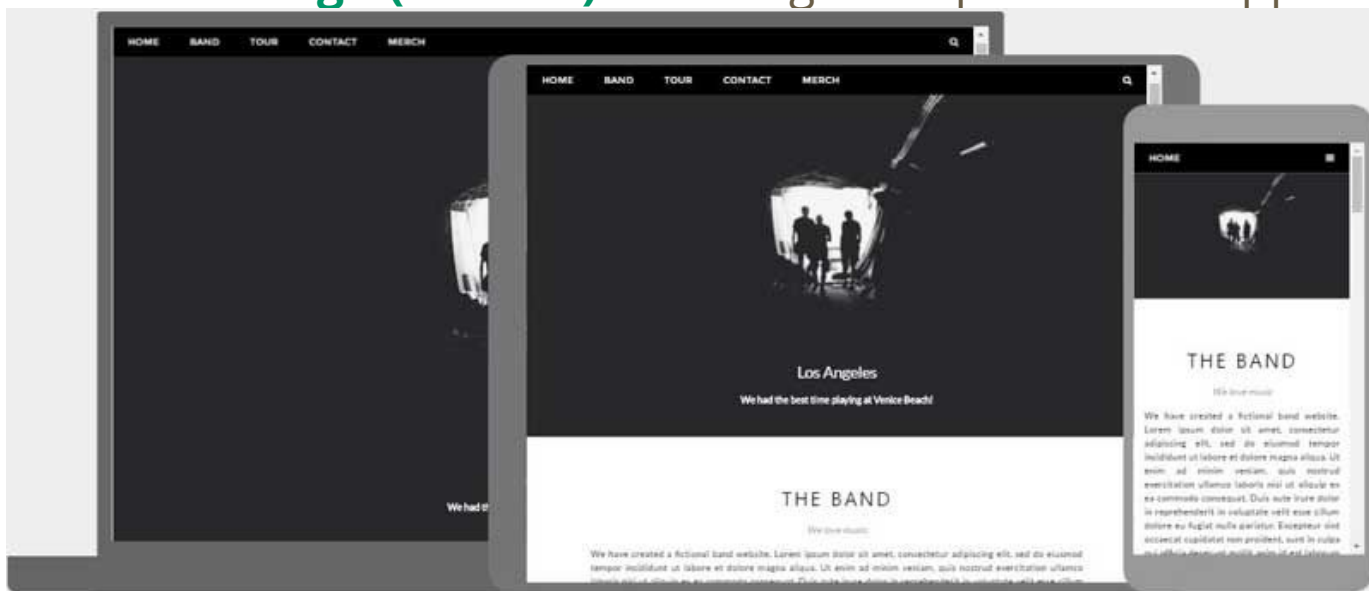
Fluid layouts

Media queries

Design strategies and patterns

Responsive Web Design

Responsive Web Design (or RWD) is a design and production approach that allows a website to adapt to different devices



Responsive Web Design

The core principle is that all devices

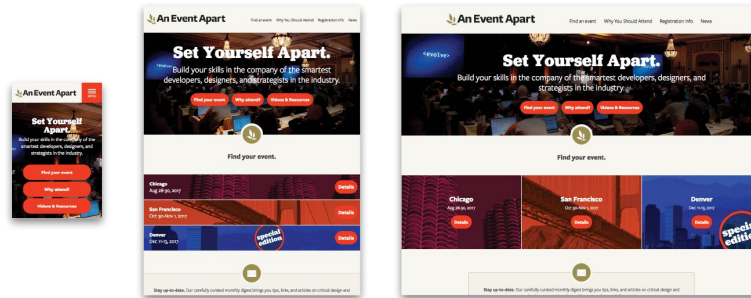
get the same HTML source,

located at the same URL, but

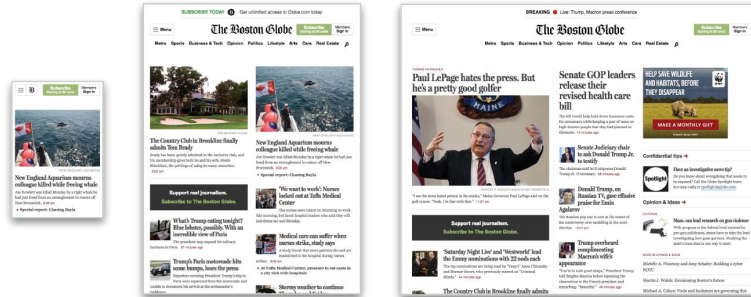
different styles are applied **based on the viewport size** to rearrange components and optimize usability

Examples are shown in the next slide

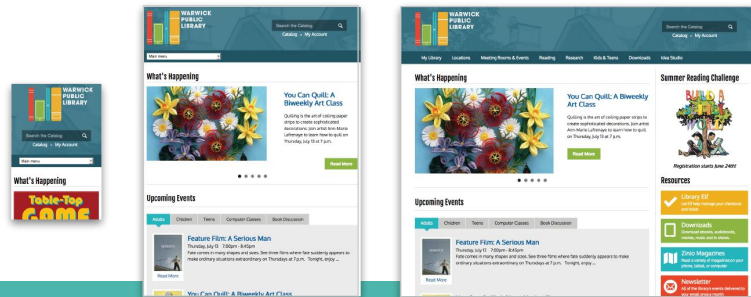
An Event Apart (aneventapart.com)



The Boston Globe (bostonglobe.com)



Warwick Public Library (warwicklibrary.org)



Techniques for RWD

A Flexible Grid

Rather than remaining at a static width, responsive sites use methods that allow them to squeeze and flow into the available browser space

Flexible Images

Images and other embedded media need to be able to scale to fit their containing elements

CSS Media Queries

Media queries give us a way to deliver sets of rules only to devices that meet certain criteria, such as width and orientation

Techniques for RWD

Setting the Viewport

To fit standard websites onto small screens, mobile browsers render the page on a canvas called the viewport and then shrink that viewport down to fit the width of the screen (device width)

Viewport

The following meta element, which goes in the head of the HTML document, tells the browser to set the width of the viewport equal to the width of the device screen (`width=device-width`)

```
<meta name="viewport" content="width=device-width,  
initial-scale=1.0">
```

With the viewport meta element in place, if the device's screen is 320 pixels wide, the rendering viewport on that device will also be 320 pixels across and will appear on the screen at full size

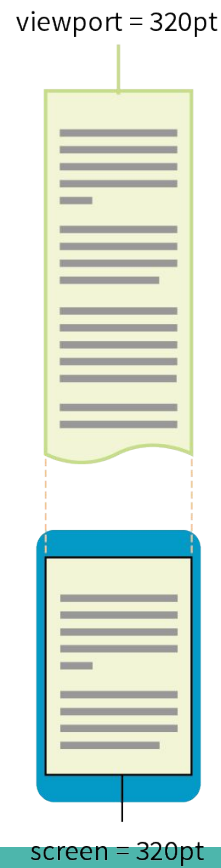
Viewport

By default, the viewport shrinks to the size of the screen.



With the viewport meta tag, the viewport is created at the same size as the screen.

```
<meta name="viewport"
      content="width=device-width,
              initial-scale=1">
```



Flexible Grids (Fluid Layouts)

In a **fluid layout**, the page area and its grid **resize proportionally** to fill the available width of the screen or window

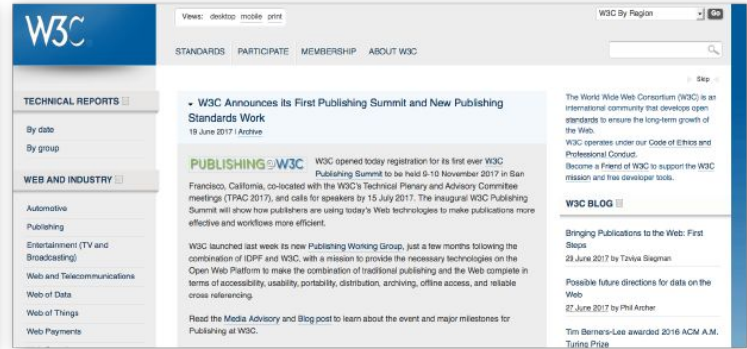
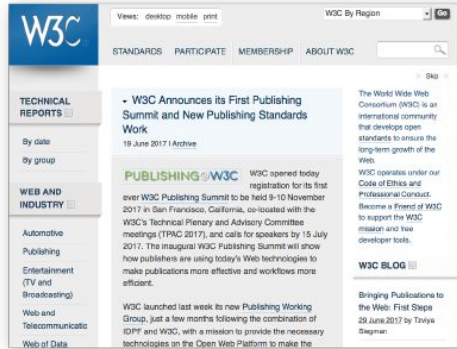
That is easily accomplished with **fr** and **minmax()** units in **CSS Grid** layouts and with **flex** property settings in **Flexbox**

Fluid layouts (or “flexible grids,”) are the foundation of responsive design

You can use percentage values for older browsers

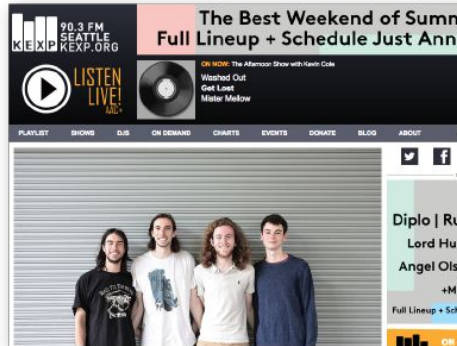
The next slide compare fixed and fluid layouts

Fluid layouts fill the viewport proportionally.



w3c.org

Fixed layouts stay the same size and may get cut off or leave extra space.



kexp.org

Making Images Flexible

If the image is larger than its container you can make the image scale down to fit the size of its container as follow

```
img {  
  max-width: 100%;  
}
```

If the container is larger than the image, the image does not scale larger than its original size

Making Images Flexible

When you apply the **max-width** property, you can omit the **width** and **height** attributes in the `img` elements in the HTML document

If you do set the **width** attribute, be sure the **height** attribute is set to **auto**

```
img {  
  max-width: 100%;  
}
```

img { max-width: 100%, }

In this book, Philadelphia Ice Creams, comprising the first group, are very palatable, but expensive. In many parts of the country it is quite difficult to get good cream. For that reason, I have given a group of creams, using part milk and part cream, but it must be remembered that it takes smart "juggling" to make ice cream from milk. By far better use condensed milk, with enough water or milk to rinse out the cans.

Ordinary fruit creams may be made with condensed milk at a cost of about fifteen cents a quart, which, of course, is cheaper than ordinary milk and cream.



If pure raw cream is stirred rapidly, it swells and becomes frothy, like the beaten whites of eggs, and is "whipped cream." To prevent this in making Philadelphia Ice Cream, one-half the cream is scalded, and when it is very cold, the remaining half of raw cream is added. This gives the smooth, light and rich consistency which makes these creams so different from others.

The time for freezing varies according to the quality of cream or milk or water; water ices require a longer time than ice creams. It is not well to freeze the mixtures too rapidly; they are apt to be coarse, not smooth, and if they are churned before the mixture is icy cold they will be greasy or "buttery."

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Media Query

Media queries apply **different styles based on characteristics of the browser** such as

- its width,

- whether it is vertically or horizontally oriented,

- its resolution

Media Query

Media Queries are what make it possible to send a **one-column layout to small screens** and a **multicolumn layout to larger screens** on the fly

Media Query

The structure of a media query as used within a stylesheet looks like the following

```
@media type and (feature: value) {  
  /* styles for browsers that meet this criteria */  
}
```

Media Query Example

The following media queries checks whether the viewport is on a screen and in landscape (horizontal) or portrait (vertical) orientation to change the background color

```
@media screen and (orientation: landscape) {  
  body { background-color: green; }  
}  
  
@media screen and (orientation: portrait) {  
  body { background-color: red; }  
}
```

Media Query Example

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```
@media screen and (orientation: landscape) {  
  body { background-color: green; }  
}  
  
@media screen and (orientation: portrait) {  
  body { background-color: red; }  
}
```

Media Types

Media Types allows to target styles to particular media

For example, the following **@media** rule delivers a set of styles only when the document is printed

```
@media print {  
  /* print-specific styles here */  
}
```

Media Types

The common media types are **all**, **print**, **screen**, and **speech**

If you are designing for **screen**, the media type is optional as shown below

```
@media (orientation: landscape) {  
  body { background-color: green; }  
}
```

```
@media (orientation: portrait) {  
  body { background-color: red; }  
}
```

Media Feature Queries

Media feature queries allow us query for a particular feature of a viewport or device

The most common feature to test for is the **viewport width**

You can also test for a minimum width (**min-width**) and maximum width (**max-width**)

Media Feature Queries Example

```
h1 {  
  font-family: Georgia, serif;  
}  
  
@media screen and (min-width: 40em) {  
  h1 {  
    font-family: "Lobster", cursive;  
  }  
}
```

List of Media Features

Feature	Description
<code>width</code>	The width of the display area (viewport). Also min-width and max-width
<code>height</code>	The height of the display area (viewport). Also min-height and max-height
<code>orientation</code>	Whether the device is in portrait or landscape orientation
<code>aspect-ratio</code>	Ratio of the viewport width divided by height (width/height)
<code>color</code>	The bit depth of the display

List of Media Features

Feature	Description
<code>color-index</code>	The number of colors in the color lookup table
<code>monochrome</code>	The number of bits per pixel in a monochrome device
<code>resolution</code>	The density of pixels in the device
<code>scan</code>	Whether a tv media type uses progressive or interlace scanning
<code>grid</code>	Whether the device uses a grid-based display, such as a terminal window

Using Media Queries

You can use media queries within a style sheet or to conditionally load external style sheets

Media queries may not be used with inline styles

Using Media Queries within a Stylesheet

The most common way to utilize media queries is to use an `@media` rule in the stylesheet

When you use media queries within a stylesheet, the order of rules is very important

rules later in the stylesheet override the rules that come before them

Using Media Queries within a Stylesheet

In RWD, the best practice in ordering rules is

- first to set up styles for small screens and browsers that don't support media queries,

- then introduce styles for increasingly larger screens later in the style sheet

Using Media Queries with External Stylesheets

For large or complicated sites, you may choose to put styles for different devices into separate stylesheets and call in the whole **.css** file when certain conditions are met

Two approaches are shown in the next two slides

Using Media Queries with External Stylesheets

One of the approach is to use the **media** attribute in the **link** element to conditionally load separate **.css** files

```
<head>
  <link rel="stylesheet" href="styles.css">
  <link rel="stylesheet" href="2column-styles.css"
media="screen and (min-width:1024px)">
</head>
```

Using Media Queries with External Stylesheets

The other approach is using `@import` rules that pull in external stylesheets from within a stylesheet

```
<style>
  @import url("/default-styles.css");
  @import url("/wide-styles.css") screen and (min-width: 1024px);
  /* other styles */
</style>
```

Choosing Breakpoints

A **breakpoint** is the point at which we use a media query to introduce a style change

Module-Based Breakpoints

Create breakpoints for the individual parts of a page

Em-Based Breakpoints

Em-based media queries keep the layout proportional to the font size

Strategies for Responsiveness

You can consider the following to optimize for responsiveness

- Content hierarchy

- Layout

- Typography

- Navigation

- Images

- Special content such as tables, forms, and interactive features

Content Hierarchy

It is best to start with an inventory of potential content and pare it down to what is most useful and important for all browsing experiences

Once you know what the content modules are, you can begin deciding in what order they appear on various screen sizes

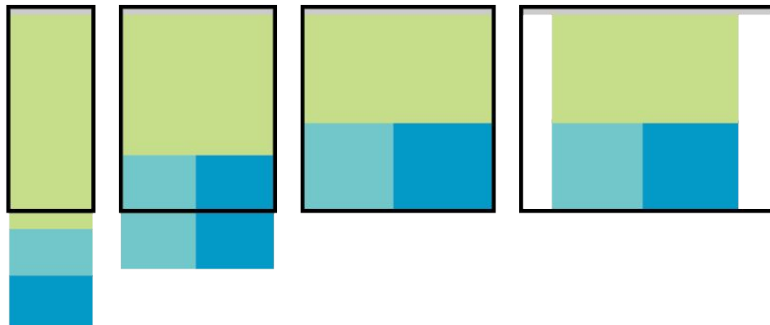
Layout

Designers typically start with a one-column layout that fits well on small handheld devices and rearrange elements into columns as more space is available

Responsive Layout Patterns

The manner in which a site transitions from a small-screen layout to a wide-screen layout must make sense for that particular site, but there are a few patterns (common and repeated approaches) that have emerged over the years

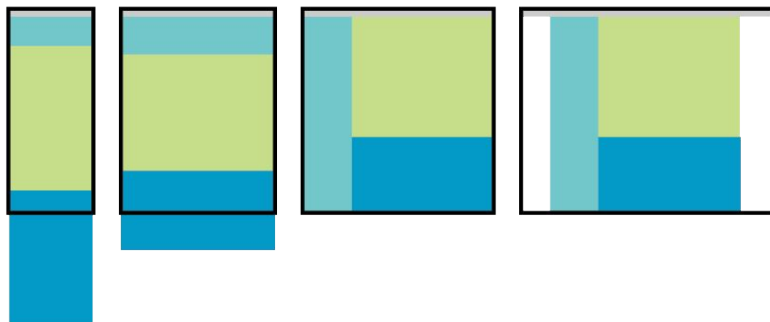
Mostly fluid



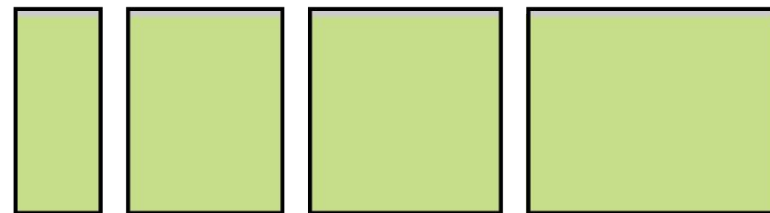
Column drop



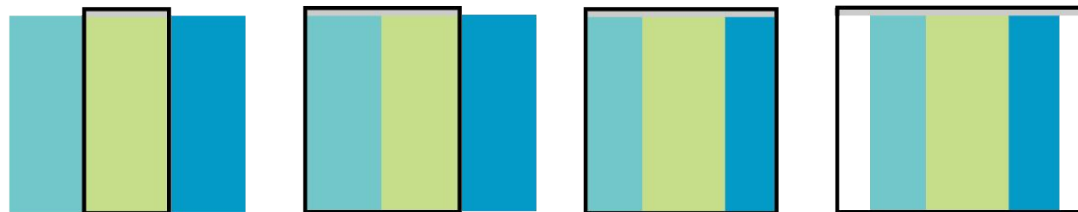
Layout shifter



Tiny tweaks



Off canvas

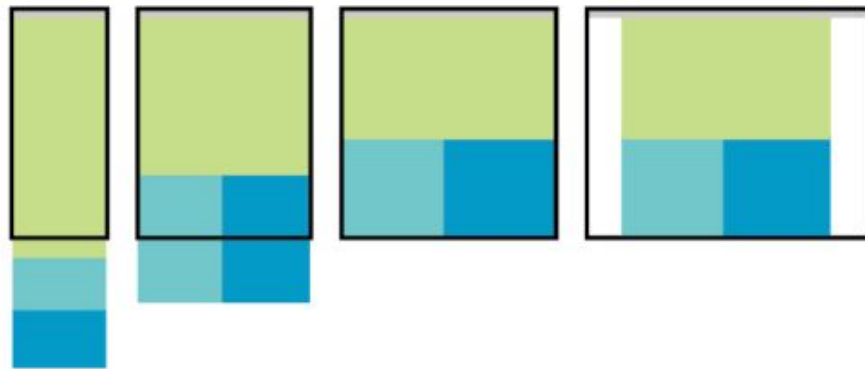


Responsive Layout Patterns

Mostly Fluid

This pattern uses a **single-column layout for small screens**, and another **fluid layout that covers medium and large screens**, with a **maximum width set to prevent it from becoming too wide**

It generally requires less work than other solutions

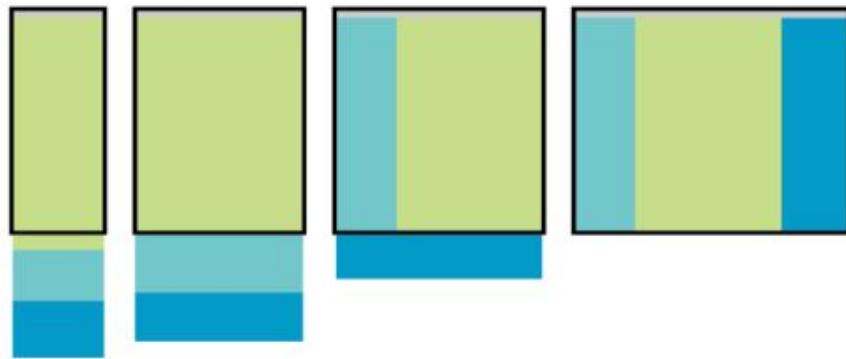


Responsive Layout Patterns

Column Drop

This solution shifts between one-, two-, and three-column layouts based on available space

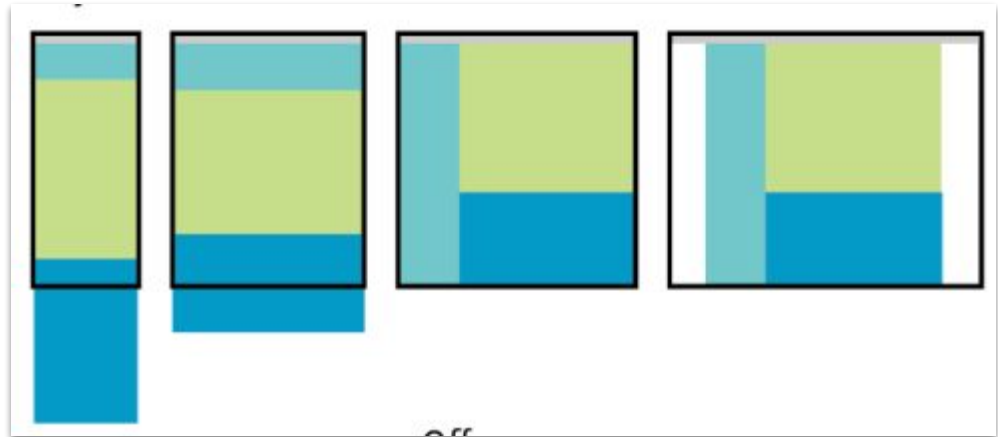
When there isn't room for extra columns, the sidebar columns drop below the other columns until everything is stacked vertically in the one-column view



Responsive Layout Patterns

Layout Shifter

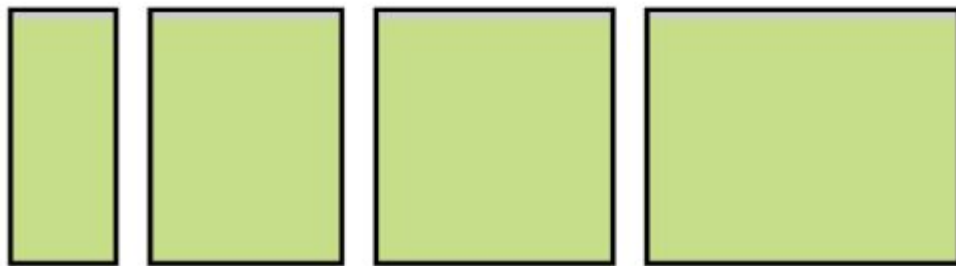
Different layout for a variety of screen sizes



Responsive Layout Patterns

Tiny Tweaks

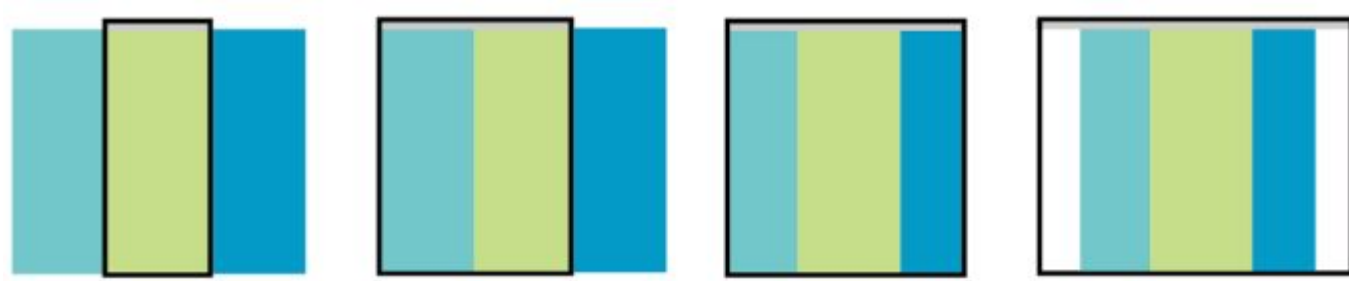
Some sites use a single-column layout and make tweaks to type, spacing, and images to make it work across a range of device sizes



Responsive Layout Patterns

Off Canvas

In this pattern, a page component is located just out of sight on the left or right of the screen and flies into view when requested



Typography

Typography requires fine-tuning along the spectrum from small-screen to wide-screen views in order to keep it legible and pleasant to read

An example is shown in the next slide

Philadelphia Ice Creams

Comprising the first group, are very palatable, but expensive. In many parts of the country it is quite difficult to get good cream. For that reason, I have given a group of creams, using part milk and part cream, but it must be remembered that it takes smart "juggling" to make ice cream from milk. By far better use condensed milk, with enough water or milk to rinse out the cans.

Ordinary fruit creams may be made with condensed milk at a cost of about fifteen cents a quart, which, of course, is cheaper than ordinary milk and cream. The cream for Philadelphia Ice Cream should be rather rich, but not double cream.

If pure raw cream is stirred rapidly, it swells and becomes frothy, like the beaten whites of eggs, and is "whipped cream." To prevent this in making Philadelphia Ice Cream, one-half

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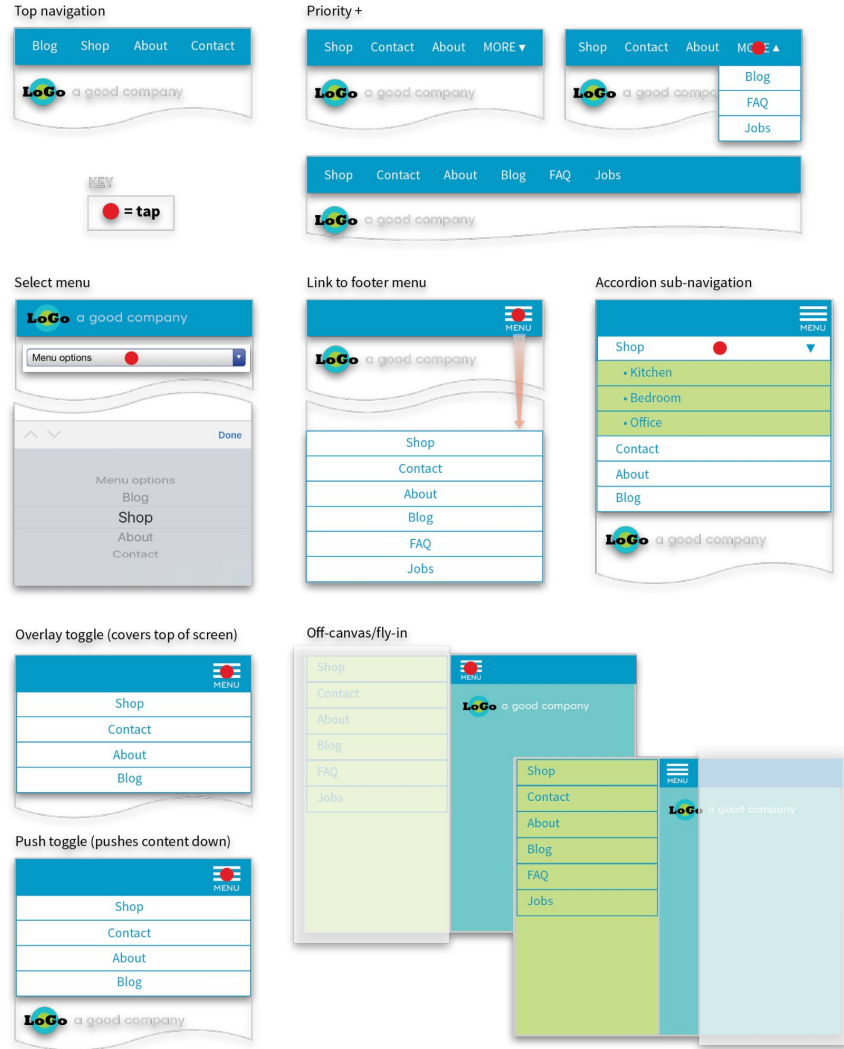
Narrow screens:

- Legible fonts
- Smaller type size
- Tighter line height
- Narrow margins

Wide screens:

- Stylized fonts OK
- Larger type size
- Open line height
- Wider margins

Navigation Patterns for small screens



Navigation

Top Navigation

If your site has just a few navigation links, they may fit just fine in one or two rows at the top of the screen

Top navigation



Navigation

Priority +

The most important navigation links appear in a line across the top of the screen alongside a More link that exposes additional options



Navigation

Link to Footer Menu

One straightforward approach places a Menu link at the top of the page that links to the full navigation located at the bottom of the page



Navigation

Accordion Sub-Navigation

Accordions that expand when you tap a small arrow icon are commonly used to reveal and hide sub-navigation



Navigation

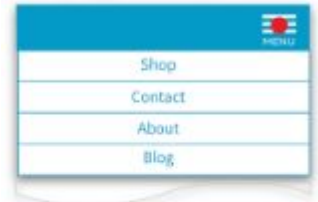
Push and Overlay Toggles

In toggle navigation, the navigation is hidden but expands downward when the menu link is tapped

It may push the main content down below it (push toggle) or slide down in front of the content (overlay toggle)



Overlay toggle (covers top of screen)



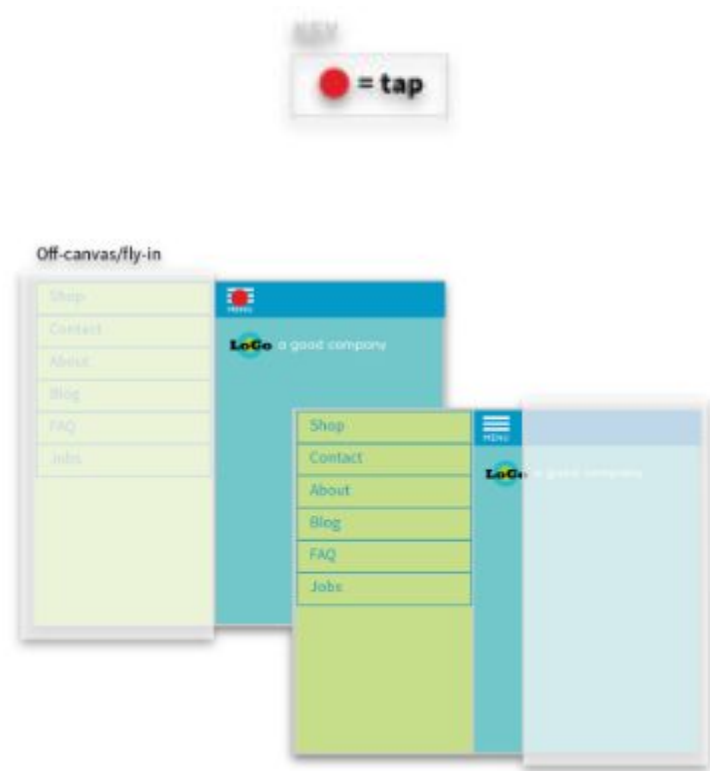
Push toggle (pushes content down)



Navigation

Off-Canvas/Fly-In

Puts the navigation in an off-screen panel to the left or right of the main content that slides into view when you tap the menu icon



Images

To make images responsive consider the following

Use responsive image markup techniques to provide multiple versions of key images for various sizes and resolutions

Read Chapter 7, Responsive Image Markup section

```
<img srcset="" src="" alt="">
```

```
<picture>  
  <source srcset="" media="" />  
  <source srcset="" media="" />  
  <img src="" alt="" />  
</picture>
```

Images

To make images responsive consider the following

Serve the smallest version as the default to prevent unnecessary data downloads

Be sure that important image detail is not lost at smaller sizes. Consider substituting a cropped version of the image for small screens

Avoid putting text in graphics, but if it is necessary, provide alternate versions with larger text for small screens

Images

A Guide to the Responsive Images Syntax in HTML

Forms

Flexbox is a great tool for adding flexibility and conditional wrapping to form fields and their labels

Tables

Responsive Data Tables

Accessible, Simple, Responsive Tables

Example

Check Next Two Slides



BLACK GOOSE BAKERY

The delicious baked goods you love at Black Goose Bistro...now available to go!

Fresh from the Oven

BREADS



Our breads are made daily from highest-quality whole grain flour, water, salt, and yeast or sourdough starter. Simply and naturally, and never any preservatives. Patience is key to achieving the proper level of fermentation and baking each loaf to perfection. Available in whole grain, sourdough, olive loaf, classic rye, and potato-onion.

[LEARN MORE ABOUT OUR BAKING PROCESS...](#)

MUFFINS



Every day, we offer a large selection of muffins, including blueberry, multi-berry, bran, corn, lemon-poppysseed, and chocolate. Our muffins are made from scratch each day. Stop by to see our seasonal muffin flavors!

[LEARN MORE ABOUT HOW WE MAKE OUR MUFFINS...](#)

Hours

MONDAY: 5am to 3pm

TUESDAY: 5am to 3pm

WEDNESDAY: 5am to 3pm

THURSDAY: 5am to 3pm

FRIDAY: 5am to 3pm

SATURDAY: 6am to 4pm

SUNDAY: 6am to 4pm

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How the Example is Made

Built using

a small-screen-first strategy,
making changes to layout,
navigation,
typography, and more at strategic breakpoints

<https://github.com/scottgruber/LWD5e/tree/master/ch17>

Further References

<https://web.dev/responsive-web-design-basics/>

[https://developer.mozilla.org/en-US/docs/Learn/CSS/CSS layout/Responsive Design](https://developer.mozilla.org/en-US/docs/Learn/CSS/CSS_layout/Responsive_Design)

https://www.w3schools.com/html/html_responsive.asp

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<https://design.tutsplus.com/articles/the-basic-elements-of-design--cms-33922>

Colors

<https://color.adobe.com/create/color-wheel>

<https://mycolor.space/>

<https://colorhunt.co/palettes/spring>

<https://colors.muz.li/color-palette-generator/800015>

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https://code.tutsplus.com/best-free-and-open-source-javascript-color-pickers--cms-37729a?_ga=2.109347973.1264349798.1699786538-2123642911.1698653606

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Typography Reference

<https://fonts.google.com/>

<https://design.tutsplus.com/articles/the-ultimate-guide-to-typography--cms-93522>

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<https://design.tutsplus.com/articles/what-is-a-sans-serif-font--cms-41957>

<https://design.tutsplus.com/articles/what-is-body-text--cms-93047>